

CS180 Project 2

Fun with Filters and Frequencies

EXPORT

SETTINGS

Fit • 100%

DEVELOP GALLERY

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PART 1: FUN WITH FILTERS

Convolutions from Scratch!

Finite Difference Operator

Derivative of Gaussian (DoG)

PART 2: FUN WITH FREQUENCIES!

Image "Sharpening"

Hybrid Images

Gaussian and Laplacian Stacks

Multiresolution Blending

QUICK ACCESS

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```
padded[cx:-cx, cy:-cy] = im1
```

```
# Convolve entire image
for r in range(padded.shape[0] - (filt.shape[0]-1)):
    for c in range(padded.shape[1] - (filt.shape[1]-1)):
        convolved[r,c] = np.sum(np.multiply(filt,
            padded[r:r+filt.shape[0], c:c+filt.shape[1]]))
```

```
return convolved
```

```
# Four-loop convolution implementation
```

```
def convolve4l(im1, filt):
```

```
    # Preprocessing (padding) implementation not shown
```

```
    # Convolve entire image with four nested loops
```

```
    for r in range(padded.shape[0] - (filt.shape[0]-1)):
        for c in range(padded.shape[1] - (filt.shape[1]-1)):
```

```
            filtered = 0
```

```
            for i in range(filt.shape[0]):
```

```
                for j in range(filt.shape[1]):
```

```
                    filtered += filt[i,j] * padded[r + i, j + c]
```

```
            convolved[r,c] = filtered
```

```
    return convolved
```

Results

Here is my selfie and a convolution with my implementation



Grayscale Selfie



Convolved Selfie

And here is a comparison to the scipy version



Scipy Convolution Result

Commentary